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# Aim: To understand DevOps, principles, practices and DevOps roles and responsibilities.

## 1. Introduction to DevOps

DevOps is a combination of **development (Dev)** and **operations (Ops)** that aims to improve collaboration, automation, and software delivery speed.

- DevOps helps organizations **deliver applications at high velocity**
- It improves **communication, collaboration, and integration**

### ◇ Problem Before DevOps

- Slow deployment (months instead of days)
- Conflicts between developers and operations
- Bugs found late in testing
- Inconsistent environments (dev, test, production)

## 2. DevOps Principles (CALMS Framework)

The **CALMS model** defines core DevOps principles:

**1. Culture**

- Focus on collaboration between teams
- Shared responsibility for success

**2. Automation**

- Reduce manual work
- Automate testing, deployment, and infrastructure

**3. Lean**

- Eliminate waste
- Improve efficiency and workflow

**4. Measurement**

- Track performance metrics like:
    - Deployment frequency
    - Failure rate

- Recovery time

**5. Sharing**

- Share knowledge across teams
- Improve communication and transparency

## 3. DevOps Practices

**1. Continuous Development**

- Planning, coding, and version control
- Frequent updates to codebase

**2. Continuous Integration (CI)**

- Code is merged frequently
- Automated builds and testing are triggered
- Benefit: Detect bugs early

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**3. Continuous Testing**

- Automated testing done repeatedly

  Ensures continuous quality improvement

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**4. Continuous Delivery & Deployment**

- **Delivery**: Code is ready for release
- **Deployment**: Code is automatically deployed

  Faster and reliable releases

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**5. Infrastructure as Code (IaC)**

- Infrastructure managed using code
- Stored in version control

  Ensures consistency across environments

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**6. Configuration Management**

- Maintains system consistency

  👉Avoids environment mismatch issues

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**7. Microservices Architecture**

- Application divided into smaller services
  Improves scalability and flexibility

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**8. Containerization**

- Use of containers (e.g., Docker)
  Ensures application runs consistently everywhere

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**9. Continuous Monitoring**

- Tracks system performance and health
  Helps detect issues early

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**10. Cloud-Based DevOps**

- Use of cloud platforms for automation and scaling
  Improves flexibility and resource usage

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## 4. DevOps Engineer Roles & Responsibilities

A DevOps Engineer acts as a **bridge between development and operations teams**.

- **Key Responsibilities:**

**1. CI/CD Pipeline Management**

- Design and maintain automated pipelines
- Ensure smooth integration and deployment

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**2. Automation**

- Automate build, testing, and deployment processes
- Reduce manual intervention

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**3. Infrastructure Management**

- Use Infrastructure as Code tools
- Manage servers and cloud resources

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**4. Monitoring & Logging**

- Monitor system performance
- Analyze logs and resolve issues

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**5. Collaboration**

- Work with developers, QA, and operations teams
- Improve communication

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**6. Security (DevSecOps)**

- Implement security practices in pipelines
- Identify vulnerabilities early

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**7. Troubleshooting**

- Identify and fix production issues
- Ensure system reliability

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**8. Version Control Management**

- Maintain code repositories (Git)
- Ensure proper versioning

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